

NORTH SEA STATION 63104, United Kingdom

WMO: 995270

Lat: 61.200N

Lon: 1.600E

Elev: 98

StdP: 14.64

Time Zone: 0.00 (EUW)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	32.2	33.9	18.4	13.9	36.4	20.8	15.7	37.6	50.4	40.1	46.5	40.8	20.2	340	0.908

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.9	65.5	59.1	62.9	58.2	61.1	57.1	60.9	63.9	59.4	62.1	58.0	60.4	11.4	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
59.6	76.4	62.1	58.1	72.4	60.6	56.7	68.9	59.4	27.0	64.1	26.1	62.4	25.2	60.5	67.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
45.4	40.4	35.6	DB	29.9	71.5	1.8	4.0	28.6	74.4	27.5	76.7	26.5	79.0	25.2	81.9
			WB	27.5	64.2	1.6	2.3	26.3	65.9	25.4	67.2	24.5	68.5	23.3	70.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.1	42.0	41.0	41.8	44.2	48.2	52.5	56.2	57.7	54.6	49.4	46.4	43.1
	DBStd	6.60	3.10	3.72	3.62	3.31	3.17	3.25	3.28	2.94	2.93	3.30	3.26	4.00
	HDD50	1392	248	252	256	176	73	11	0	0	3	50	110	213
	HDD65	6160	713	672	721	624	520	376	275	229	312	484	557	677
	CDD50	707	0	0	0	2	18	84	191	237	141	31	3	0
	CDD65	3	0	0	0	0	0	0	2	1	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	18.5	23.3	19.9	19.6	17.7	15.5	14.5	14.9	14.8	17.3	19.9	21.4	22.6
Precipitation	PrecAvg	52.8	6.3	5.2	5.0	2.8	1.9	2.6	3.2	3.9	4.6	5.6	5.5	6.1
	PrecMax	65.0	9.9	10.1	9.3	4.8	3.3	5.2	6.0	6.4	7.5	8.5	8.7	9.4
	PrecMin	37.6	3.0	2.3	1.8	1.0	0.9	1.0	1.6	2.6	1.9	2.3	1.7	2.6
	PrecStd	6.5	1.9	2.1	1.8	0.9	0.6	0.9	0.9	1.0	1.3	1.6	1.5	1.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.3	49.5	49.9	56.0	59.9	66.0	69.7	69.8	64.2	57.6	53.3	51.0
		MCWB	46.3	46.9	46.3	48.6	52.3	56.5	61.4	60.6	58.2	53.3	50.6	48.6
	2%	DB	48.3	48.0	48.3	51.5	56.5	61.5	65.6	65.3	60.7	55.6	52.1	50.1
		MCWB	45.7	45.5	45.4	46.3	50.3	55.0	59.7	59.3	57.0	51.8	49.5	48.0
	5%	DB	47.5	47.1	47.3	49.4	54.1	58.8	62.5	63.1	59.1	54.4	51.2	49.4
		MCWB	44.8	44.7	44.4	45.5	48.8	53.7	58.2	58.5	56.4	50.8	48.7	47.2
	10%	DB	46.6	46.1	46.3	48.0	52.2	56.5	60.6	61.4	58.0	53.3	50.4	48.5
		MCWB	43.8	43.7	43.5	45.0	48.1	52.1	57.1	57.5	55.7	49.8	48.0	46.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.7	48.1	47.7	49.6	54.1	59.3	63.0	62.6	61.0	55.3	51.9	50.1
		MCDB	48.2	48.7	48.7	54.4	58.7	63.6	67.3	66.1	61.8	56.5	52.4	50.2
	2%	WB	46.6	46.7	46.5	47.6	51.5	55.9	60.7	61.1	58.6	53.1	50.7	48.9
		MCDB	47.6	47.4	47.6	49.7	54.3	60.3	63.8	64.2	59.8	54.3	51.4	49.7
	5%	WB	45.5	45.4	45.3	46.6	50.2	54.3	59.2	59.8	57.4	51.5	49.8	47.7
		MCDB	46.9	46.5	46.7	48.6	52.8	57.9	62.5	62.5	58.3	53.1	50.7	49.0
	10%	WB	44.2	43.9	44.1	45.6	49.1	53.0	57.5	58.3	56.4	50.5	48.9	46.5
		MCDB	46.1	45.6	45.9	47.7	51.5	55.8	60.2	60.7	57.7	52.5	50.0	48.2
Mean Daily Temperature Range	5% DB	MDBR	4.9	5.1	5.3	5.7	6.3	6.0	5.6	5.9	4.8	4.7	4.5	5.0
		MCDBR	5.5	5.9	6.0	7.7	9.1	9.4	8.5	8.8	6.4	5.4	4.8	4.9
		MCWBR	6.3	5.7	5.8	5.7	6.1	5.9	5.1	5.6	4.9	5.0	5.1	5.8
	5% WB	MCDBR	5.3	4.9	5.3	6.8	7.7	8.1	7.7	7.5	5.3	4.6	4.7	4.9
		MCWBR	6.7	6.0	6.2	5.7	5.7	5.6	4.8	5.6	4.5	5.2	5.6	6.4
Clear-Sky Solar Irradiance	taub	0.334	0.355	0.368	0.380	0.369	0.361	0.380	0.369	0.364	0.345	0.319	0.380	
	taud	2.062	2.185	2.267	2.306	2.409	2.472	2.437	2.466	2.431	2.308	2.128	2.290	
	Ebn at Noon	108	188	234	257	271	275	266	259	237	193	114	68	
	Edh at Noon	14	23	29	34	33	31	32	29	25	20	13	9	
All-Sky Solar Radiation	RadAvg	60	213	570	1112	1552	1588	1440	1151	697	326	94	37	
	RadStd	8	26	63	90	112	173	128	107	56	41	12	2	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (22 neighbors)	N/A	+1.17	+1.12	N/A	N/A	N/A	-184	-241	N/A	N/A

Nomenclature: See separate page